

1. Prove convergence and calculate the integral

$$\int_0^{+\infty} x^3 e^{-x} dx.$$

2. Prove convergence and find the sum of the series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k(k+1)}.$$

3. Check the convergence of the series

$$\sum_{k=1}^{\infty} \sin \frac{(\sqrt{k}+1)^3}{k^3+2k+1}.$$

4. Check the convergence of the improper integral

$$\int_0^{\infty} \frac{\sin \sqrt{x}}{x} dx.$$

5. Find the length of a path

$$x = \sin t, \quad y = \sin 2t, \quad t \in [0, 2\pi].$$

6. Find the radius of convergence and the sum of the power series

$$f(x) = \sum_{k=1}^{\infty} \frac{2k}{2k+1} x^{2k+1}.$$

7. Calculate the integral

$$\int_0^1 \ln(1-x) dx.$$

8. Let $p > 0$. Prove that

$$\lim_{n \rightarrow \infty} \int_n^{n+p} \frac{\sin^2 x}{x} dx = 0.$$

9. Determine without evaluating which integral is greater

$$\int_0^{\pi} e^{-x^2} \sin^2 x dx \quad \text{or} \quad \int_{\pi}^{2\pi} e^{-x^2} \sin^2 x dx.$$

10. Check the uniform convergence of the series

$$\sum_{k=1}^{\infty} \frac{1}{n} \ln \frac{n+x}{n}$$

on $E_1 = (0, 1)$ and $E_2 = (1, +\infty)$.

11. Prove the D'Alembert's ratio test for convergence of a series.

12. Check convergence and absolute convergence of a series

$$\sum_{k=1}^{\infty} \frac{\sin^3 k}{k}.$$

13. Find $\lim x_n$, where

$$x_n = \frac{1}{\sqrt{4n^2 - 1^2}} + \frac{1}{\sqrt{4n^2 - 2^2}} + \dots + \frac{1}{\sqrt{4n^2 - n^2}}.$$

14. Prove that the absolutely convergent numerical series is convergent.

15. Prove the comparison test for improper integrals.

16. Prove the formula for integration by parts for Riemann integral.

17. Find the area of a curvilinear sector bounded by a curve defined in polar coordinates

$$r(t) = 2 + \cos t, \quad t \in [0, 2\pi].$$